

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EC-601 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Industrial Electronics
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Draw the circuit diagram of an single phase half wave controlled rectifier with R-L load. Derive the expression for average output voltage (D.C)
2. Compare between thyristor and transistor.
3. Draw and explain the circuit diagram of shunt voltage regulator. Give some of the differences between the fixed and adjustable voltage regulators.
4. Four thyristors each of which passes rating of 1000V are connected in series. The total voltage rating of the string is 3200V. What is the value of the string efficiency.
5. a) What do you mean by Ladder diagram?
b) What do you mean by interface module. List its functions in a given PLC.
6. Why do we need parallel operation of an SCR? What are the difficulties observed while using parallel operation?

OR

Explain the constructional details and working of the enhancement type power MOSFET. Why VMOS/UMOS structures able to withstand higher power rating.

7. Draw the circuit diagram of a series pass transistor voltage regulator.

OR

Classify the IGBT structures available in the market and discuss any one structure along with its characteristics.

8. Write short notes (any two)
 - a) PLC
 - b) Power supplies using OPAMP
 - c) Ladder Logic Language
 - d) Microprocessor in PLC
